# Continuous-time embedding algorithms for absorbing Markov chains.

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**Abstract**

Continuous-time Markov chain (CTMC) models are useful in a wide range of disciplines such as engineering, medical sciences, and finance. Embedding algorithms conveniently provide a continuous-time approximation in case only a discrete-time model is available, but typically rely on generic loss functions that prioritize single-step transitions and treat all states as equally important. We argue that this ignores the importance of multi-step survival probabilities in absorbing Markov chains. In this work, we propose a generalized framework for constructing Expectation-Maximization (EM) algorithms that minimize divergence-based embedding loss functions. Using this framework, we develop a specialized EM algorithm designed to better preserve the original probability distribution of survival times. Numerical experiments based on empirical credit rating transition matrices show that our approach strikes a robust balance in the trade-off between preserving survival probabilities and fitting overall transition dynamics across different time horizons.

*Keywords:*
embedding problem, absorbing Markov chain, survival probabilities, expectation-maximization, Kullback-Leibler divergence

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## 1. Introduction

Markov chains are widely used for modeling stochastic state transitions in disciplines such as engineering, medical sciences, and finance. Although a continuous-time Markov chain (CTMC) model conveniently considers state transitions at any future time $t \geq 0$, sometimes only a discrete-time Markov chain (DTMC) is available, represented by a transition matrix. This limitation could arise from discrete observation frequencies, such as periodic maintenance checks in engineering, or from a third-party providing the transition matrix, such as annual publications by credit rating agencies. In this work, we propose a framework for approximate continuous-time embedding, with a focus on preserving discrete-time information about survival times in absorbing Markov chains.

Given a DTMC with transition matrix $P$, the well-known (continuous-time) embedding problem consists of finding a CTMC that matches it at discrete time points (Elfving, 1937; Kingman, 1962). This is equivalent to finding an infinitesimal generator matrix $Q$ that satisfies $e^Q = P$. However, such a generator often does not exist and may not be unique (Israel et al., 2001). Even when a real matrix logarithm of $P$ exists, it often fails to satisfy the necessary conditions of a generator matrix, such as having non-negative off-diagonal elements. As of this writing, necessary and sufficient criteria for the existence of a continuous-time embedding have only been established for Markov chains with up to 4 states (Casanellas et al., 2023; Baake and Sumner, 2024). However, some necessary conditions are known for arbitrary dimensions (see e.g. Israel et al., 2001; Davies, 2010). For example, an embeddable transition matrix $P$ should satisfy $P_{kl} > 0$ if $P^n_{kl} > 0$ for some $n \in \mathbb{N}$. This condition is often violated in empirical transition matrices, where the probability of rare unobserved state transitions is incorrectly estimated as zero. Kingman (1962) proved that a continuous-time embedding exists if and only if there exists a stochastic $p$th root of $P$ for all $p \in \mathbb{N}$, thus connecting it to the discrete-time embedding problem of finding stochastic roots of transition matrices (see e.g. Lin, 2011; Van-Brunt, 2018; Camellini et al., 2025; Joshi and Šmigoc, 2025).

For a non-embeddable DTMC, one can choose an approximate embedding that matches the available discrete-time information as closely as possible. For example, some propose an approximate embedding based on the additional assumption of at most a single jump per discrete time step (Jarrow et al., 1997; Carette and Guerry, 2025). Others propose regularization approaches, which start with a matrix logarithm of the transition matrix $P$ and modify it to obtain a valid generator matrix (see e.g. Israel et al., 2001; Kreinin and Sidelnikova, 2001; Hughes and Werner, 2016). A more structured approach consists of first defining an embedding loss function $\mathrm{Loss}(Q, P)$ and then constructing an algorithm to minimize it with respect to $Q$. For example, a common choice is the $L^2$-norm distance between the transition matrices $e^Q$ and $P$ (see e.g. Israel et al., 2001; Kreinin and Sidelnikova, 2001; He and Gunn, 2003; Lin, 2011; Hughes and Werner, 2016; Durastante and Meini, 2024), which corresponds to the problem of finding the "nearest" embeddable transition matrix (see e.g. Higham, 1989).

Stokes (2024) instead proposes divergence-based embedding loss functions, which are more natural in the context of approximating a stochastic model. As we highlight, Expectation-Maximization (EM) embedding algorithms (see e.g. Inamura, 2006; Dos Reis and Smith, 2018) implicitly minimize the Kullback Leibler (KL) divergence loss considered by Stokes (2024). This improves on the $L^2$-norm by naturally weighing errors relative to the original probability mass, but still treats all single-step state transitions as equally important. In contrast, we argue that absorbing states often represent rare critical events, such as a machine failure in systems engineering, the death of a patient in medical sciences, or the bankruptcy of an obligor in credit risk management. For example, modeling credit rating transitions with an absorbing default state is a prominently mentioned application in embedding algorithm literature (see e.g. Israel et al., 2001; Kreinin and Sidelnikova, 2001; Inamura, 2006; Lin, 2011; Hughes and Werner, 2016; Dos Reis and Smith, 2018; Stokes, 2024).

In this work, we propose a framework for EM embedding algorithms based on KL divergence loss functions, which can specify the discrete-time information that should be preserved. Our framework provides a unifying generalization of existing methods with new matrix-form expressions for the expectation step. Using our framework, we construct a new EM algorithm for absorbing Markov chains that targets the distribution of survival times and single-step transitions. Through numerical experiments on empirical credit rating transition matrices, we showcase that our algorithm outperforms existing approaches in preserving discrete-time survival probabilities, while maintaining a reasonable fit to overall single-step state transitions. We compare the performance of the divergence-based algorithms with regularization approaches and $L^2$-norm alternatives, for which we derive a closed-form expression for the matrix gradient.

Section 2 outlines our framework for constructing embedding algorithms, Section 3 presents the numerical results, and Section 4 concludes. Appendix A and Appendix B outline technical details that were omitted from the main text.

## 2. Framework for embedding algorithms

Before we outline our framework for constructing embedding algorithms based on divergence-based loss functions, we first introduce some notation.

### 2.1. Terminology and notation

Throughout this work, we consider a time-homogeneous DTMC $\mathrm{D} = \{\mathrm{D}_n\}_{n\in\mathbb{N}_0}$ on a finite state space $E = \{1, \ldots, K\}$, where $\mathrm{D}_n$ represents the state at time $n$. We consider a fixed probability distribution $\pi = (\pi_k)_{k\in E}$ for the initial state $\mathrm{D}_0$. For any probability measure $\mathbb{P}$ on the set $\Omega = E^{\mathbb{N}}$ of all possible DTMC trajectories, we denote the probability measure conditional on the initial state $\mathrm{D}_0 = k$ as $\mathbb{P}_k$. Given $\pi$, a DTMC is characterized by a transition probability matrix $P \in \mathcal{P}$, where $\mathcal{P}$ denotes the set of all stochastic matrices of dimension $K$. That is $\mathcal{P} := \left\{P \in \mathbb{R}^{K\times K} \mid P\mathbb{1} = \mathbb{1}; P_{kl} \geq 0,\ \forall k, l \in E\right\}$, where $\mathbb{1}$ is a $K$-dimensional vector with all elements equal to 1. The element $P_{kl}$ represents the probability of transitioning from state $k$ to state $l$ in a single time step. We denote $\mathbb{P}^P$ for the probability measure on $\Omega$ that is induced by the transition matrix $P$ and initial distribution $\pi$. For example, we have $\mathbb{P}^P_k[\mathrm{D}_1 = l] = P_{kl}$.

Similarly, a continuous-time Markov chain (CTMC) $\mathrm{C} = \{\mathrm{C}_t\}_{t\geq 0}$ on $E$ is characterized by initial distribution $\pi$ and an (infinitesimal) generator matrix $Q \in \mathcal{Q}$, where $\mathcal{Q} := \left\{Q \in \mathbb{R}^{K\times K} \mid Q\mathbb{1} = 0; Q_{kl} \geq 0,\ \forall k \neq l\right\}$. By discretely sampling the CTMC C, we obtain a DTMC that equals $\mathrm{C}_n$ for all $n \in \mathbb{N}$ and has transition matrix $e^Q := \sum_{n=0}^{\infty} \frac{1}{n!} A^n$. We denote $\mathbb{P}^Q$ for the probability measure on $\Omega$ induced by the CTMC with generator $Q$ and initial distribution $\pi$. For example, we have $\mathbb{P}^Q_k[\mathrm{D}_1 = l] = [e^Q]_{kl}$.

In the next section, we consider embedding algorithms based on loss functions that generalize the standard Kullback-Leibler divergence (Stokes, 2024)

$$\mathrm{Loss}^{\mathrm{Standard}}_{\mathrm{KL}}(Q, P) = \sum_{k,l\in E} P_{kl} \log\left(\frac{P_{kl}}{[e^Q]_{kl}}\right). \tag{1}$$

For our proposed algorithms that preserve survival times, we consider an absorbing Markov chain with a single absorbing state $K$. The (discrete) survival time $\tau$ represents the number of discrete time steps before reaching the absorbing state, i.e. $\tau := \inf\{n \in \mathbb{N} : \mathrm{D}_n = K\}$. For any $n \in \mathbb{N}$, the conditional survival probabilities are computed as $\mathbb{P}^P_k[\tau \leq n] = u_k^\top P^n u_K$ and $\mathbb{P}^Q_k[\tau \leq n] = u_k^\top e^{Qn} u_K$, where $u_l$ is the standard $K$-dimensional unit vector with a 1 in the $l$th element.

### *2.2. Divergence-based embedding algorithms*

Divergence-based embedding algorithms essentially (implicitly) aim to preserve the probability distribution of a chosen random variable $Y : \Omega \to \mathcal{Y}$, which maps the discrete trajectories $\Omega$ of the DTMC to a countable outcome space $\mathcal{Y}$. Conceptually, the choice of $Y$ can be interpreted as specifying which probabilistic discrete-time information is important. We focus on KL divergence due to the natural link to likelihood maximization and EM algorithms.

#### *2.2.1. Kullback-Leibler divergence loss functions*

We consider a general KL divergence embedding loss function of the form

$$\mathrm{Loss}^{Y}_{\mathrm{KL}}(Q, P) = D^{Y}_{\mathrm{KL}}(\mathbb{P}^P \| \mathbb{P}^Q) := \sum_{y\in\mathcal{Y}} \mathbb{P}^P(Y = y) \log\left(\frac{\mathbb{P}^P(Y = y)}{\mathbb{P}^Q(Y = y)}\right),$$

where we use the conventions $0 \log(0/x) = 0$ for all $x \geq 0$ and $z \log(z/0) = \infty$ for all $z > 0$. The loss function quantifies how much the probability distribution of $Y$ differs under the approximate continuous-time embedding $\mathbb{P}^Q$, relative to the "true" original discrete-time model $\mathbb{P}^P$.

For a target variable of the form $Y = [\mathrm{D}_0, \tilde{Y}]$, we have outcome space $\mathcal{Y} = E \times \tilde{\mathcal{Y}}$ and decomposition $D^{[\mathrm{D}_0,\tilde{Y}]}_{\mathrm{KL}}(\mathbb{P}^P \| \mathbb{P}^Q) = \sum_{k\in E} \pi_k D^{\tilde{Y}}_{\mathrm{KL}}(\mathbb{P}^P_k \| \mathbb{P}^Q_k)$. Conceptually, this loss measures the error in the distribution of $\tilde{Y}$ conditional on each possible initial state, weighted by the initial distribution $\pi$ For example, the choice $Y = [\mathrm{D}_0, \mathrm{D}_n]$ corresponds to preserving the conditional dynamics of $n$-step state transitions. In this case, we have $\mathcal{Y} = E \times E$ and

$$\mathrm{Loss}_{\mathrm{KL}}^{[\mathrm{D}_0,\mathrm{D}_n]}(Q, P) = \sum_{k,l \in E} \pi_k [P^n]_{kl} \log\left(\frac{[P^n]_{kl}}{[e^{Qn}]_{kl}}\right). \tag{2}$$

In particular, the loss $\mathrm{Loss}_{\mathrm{KL}}^{\mathrm{Standard}}$ considered by Stokes (2024) corresponds to $n = 1$ and a uniform initial distribution $\pi_k \equiv \frac{1}{K}$, up to a multiplicative constant. In Section 3, we consider $n = 5, 10, 20$ to assess the performance of embedding algorithms over longer time horizons.

For our motivating example of absorbing Markov chains, we consider the special case $Y = [\mathrm{D}_0, \tau]$, corresponding to the combination of initial state and survival times. In this case, the outcome space equals $\mathcal{Y} = E \times \mathbb{N}$ and $\mathrm{Loss}_{\mathrm{KL}}^{\mathrm{Survival}} \equiv \mathrm{Loss}_{\mathrm{KL}}^{[\mathrm{D}_0,\tau]}$ equals

$$\mathrm{Loss}_{\mathrm{KL}}^{\mathrm{Survival}}(Q, P) = \sum_{k \in E} \pi_k \sum_{n=1}^{\infty} \mathbb{P}_k^P\big(\tau = n\big) \log\left(\frac{\mathbb{P}_k^P\big(\tau = n\big)}{\mathbb{P}_k^Q\big(\tau = n\big)}\right). \tag{3}$$

This loss measures the embedding error in the conditional distribution of survival times in absorbing Markov chains. A conceptually similar divergence-based transition matrix loss function was considered by Gzyl and Mayoral (2025).

Finally, for two target variables $Y_1, Y_2$ that are independent under $\mathbb{P}^P$ and $\mathbb{P}^Q$, we have that $\mathrm{Loss}_{\mathrm{KL}}^{Y_1,Y_2}(Q, P) = \mathrm{Loss}_{\mathrm{KL}}^{Y_1}(Q, P) + \mathrm{Loss}_{\mathrm{KL}}^{Y_2}(Q, P)$. Intuitively, this corresponds to observing the target variables on mutually independent trajectories of the DTMC. This decomposition can be used to construct joint loss functions that aim to preserve multiple characteristics of the original DTMC. In particular, we consider a loss that measures the error in the conditional distribution of both survival times and single-step transitions

$$\mathrm{Loss}_{\mathrm{KL}}^{\mathrm{Enhanced}}(Q, P) := \mathrm{Loss}_{\mathrm{KL}}^{\mathrm{Standard}}(Q, P) + \mathrm{Loss}_{\mathrm{KL}}^{\mathrm{Survival}}(Q, P). \tag{4}$$

#### *2.2.2. EM-based embedding algorithms*

We highlight a well-known link between minimizing KL divergence with respect to a known distribution and maximizing expected likelihood through EM algorithms (see e.g. Bladt and Nielsen, 2017, pp. 684–685). More specifically, $\arg\min_Q D_{\mathrm{KL}}^Y(\mathbb{P}^P \| \mathbb{P}^Q) = \arg\max_Q \mathbb{E}^P\left[\ell\left(Q \mid Y\right)\right]$, where $\ell\left(Q \mid y\right)$ represents the log-likelihood of a single observation $y$ for the CTMC with generator $Q$, and where $\mathbb{E}^P$ represents the expectation under the probability measure $\mathbb{P}^P$. This expectation essentially corresponds to the log-likelihood of an infinite i.i.d. sample of observations $Y^{(m)}$ under $\mathbb{P}^P$, since

$$\frac{1}{M}\ell\big(Q \mid \{Y^{(m)}\}_{m=1}^M\big) = \frac{1}{M}\sum_{m=1}^{M} \ell\big(Q \mid Y^{(m)}\big) \xrightarrow[\mathbb{P}^P\text{-a.s.}]{M\to\infty} \mathbb{E}^P\left[\ell\left(Q \mid Y\right)\right].$$

Although gradient-based optimization of the log-likelihood has been considered in the context of CTMCs (see e.g. Kalbfleisch and Lawless, 1985), we adopt the more common EM algorithm approach.

Maximizing the (log-)likelihood of a fully observed CTMC is straightforward through the sufficient statistics $J = [J_{kl}]_{k,l\in E}, T = [T_k]_{k\in E}$, where $J_{kl}$ represents the total number of jumps from state $k$ to $l$ and $T_k$ represents the total time spent in state $k$. More specifically, the log-likelihood attains its maximum for the generator $Q$ that satisfies $Q_{kl} = J_{kl}/T_k$ for all $k \neq l$ (see Albert, 1962).

In this perspective, the discrete target variable $Y$ essentially represents "incomplete data" relative to the entire continuous-time path. Therefore, a natural optimization approach is to use an EM algorithm, which switches between an expectation-step (E-step) and a maximization-step (M-step) to iteratively update the generator (see e.g. Dempster et al., 1977; Wu, 1983; McLachlan and Krishnan, 2008). In our context, the E-step computes the expectations $\widehat{J} := \mathbb{E}^P\big[\mathbb{E}^Q\big[J \mid Y\big]\big], \widehat{T} := \mathbb{E}^P\big[\mathbb{E}^Q\big[T \mid Y\big]\big]$, and the M-step updates the generator by $Q_{kl} = \widehat{J}_{kl}/\widehat{T}_k$ for all $k \neq l$ (see e.g. Dos Reis and Smith, 2018). Algorithm 1 in Appendix B outlines the full EM embedding algorithm.

We note that EM algorithms for discretely-observed CTMCs are often considered for continuous-time embedding (e.g. Inamura, 2006; Bladt and Sørensen, 2005, 2009; Dos Reis and Smith, 2018). In this context, these algorithms essentially correspond to the special case of $Y = [\mathrm{D}_0, \mathrm{D}_1]$, and thus (implicitly) minimize the loss $\mathrm{Loss}_{\mathrm{KL}}^{\mathrm{Standard}}$ considered by Stokes (2024). Such EM algorithms mostly differ in how they compute the E-step. Dos Reis and Smith (2018) propose using closed-form element-wise expressions for the E-step, as this yields faster and more accurate results than the MCMC-based EM algorithms considered in Bladt and Sørensen (2005); Inamura (2006); Bladt and Sørensen (2009).

We adopt a similar approach and derive matrix-form expressions for the E-step using matrix exponential integral results by Van Loan (1978). Proposition 1 in Appendix A.1 provides an efficient expression for the E-step of the special case $Y = [\mathrm{D}_0, \mathrm{D}_1]$, which requires only a single matrix exponential evaluation instead of one for each element. Using similar techniques, Proposition 2 in Appendix A.1 provides a new matrix-form expression for the E-step corresponding to the special case $Y = [\mathrm{D}_0, \tau]$. We note that this result is conceptually similar to existing EM algorithms based on censored phase-type distributions (see e.g. Olsson, 1996; Bladt and Nielsen, 2017). To minimize the combined loss $\mathrm{Loss}_{\mathrm{KL}}^{\mathrm{Enhanced}}$ in Equation (4), we propose an Enhanced EM algorithm that simply sums the two E-step results corresponding to $Y = [\mathrm{D}_0, \mathrm{D}_1]$ and $Y = [\mathrm{D}_0, \tau]$.

### *2.3. $L^2$-norm embedding approaches*

To benchmark the divergence-based approaches, we consider two $L^2$-norm alternatives: the existing Standard $L^2$ algorithm and a new Enhanced $L^2$ algorithm. The Standard $L^2$ algorithm minimizes the distance between the original transition matrix $P$ and the embedded $e^Q$

$$\mathrm{Loss}_{L^2}^{\mathrm{Standard}}(Q, P) := \left\|e^Q - P\right\|_{L^2}^2, \quad \text{where } \|A\|_{L^2}^2 = \mathrm{Tr}(A^\top A), \tag{5}$$

which is commonly used for evaluating a continuous-time embedding (see e.g. Israel et al., 2001; Kreinin and Sidelnikova, 2001; He and Gunn, 2003; Lin, 2011; Hughes and Werner, 2016; Durastante and Meini, 2024) or a discrete-time embedding (see e.g. He and Gunn, 2003; Lin, 2011; Durastante and Meini, 2024; Camellini et al., 2025). This loss measures an overall fit over single-step state transition probabilities, analogous to the standard KL divergence loss $\mathrm{Loss}_{\mathrm{KL}}^{\mathrm{Standard}}$ defined in Equation (1).

For the new Enhanced $L^2$ algorithm, we consider the error in the term structure of conditional survival probabilities, weighted by the initial distribution

$$\mathrm{Loss}_{L^2}^{\mathrm{Survival}}(Q, P) = \sum_{k \in E} \pi_k \sum_{n=1}^{\infty} \left(\mathbb{P}_k^Q[\tau \leq n] - \mathbb{P}_k^P[\tau \leq n]\right)^2. \tag{6}$$

This loss measures the overall fit in survival probabilities over longer time horizons, analogous to the KL divergence loss $\mathrm{Loss}_{\mathrm{KL}}^{\mathrm{Survival}}$ defined in Equation (3). We note that a conceptually similar transition matrix loss was considered by J.P. Morgan (1997, pp. 70–72). Analogous to our Enhanced EM algorithm, our Enhanced $L^2$ algorithm minimizes the combination $\mathrm{Loss}_{L^2}^{\mathrm{Standard}} + \mathrm{Loss}_{L^2}^{\mathrm{Survival}}$.

Proposition 3 in Appendix A.2 derives a closed-form expression of the matrix gradient for efficient numerical optimization. We note that the Standard $L^2$ algorithm essentially corresponds to a gradient-based variant of the BAM method proposed by Hughes and Werner (2016).

## 3. Numerical results

We compare the performance of the embedding algorithms on empirical credit rating transition matrices with absorbing default states. We show that our Enhanced EM algorithm outperforms existing methods in preserving survival probabilities, while maintaining a reasonable fit over single-step transitions.

### *3.1. Set-up*

Before presenting the results, we briefly outline the embedding algorithms and the empirical transition matrices that are considered.

#### *3.1.1. Compared embedding algorithms*

The main goal is to compare the EM-based embedding algorithms described in Section 2.2: our Enhanced EM algorithm and the existing Standard EM algorithm, which essentially corresponds to the approach by Dos Reis and Smith (2018). Additionally, we consider the two alternatives described in Section 2.3: our Enhanced $L^2$ algorithm and the existing Standard $L^2$ algorithm. Finally, we also consider the three existing regularization methods described in Appendix B.2: the Diagonal Adjustment (DA) and Weighted Adjustment (WA) methods in Equations (B.1) and (B.2) (Israel et al., 2001), as well as the Quasi-Optimization of the Generator (QOG) method in Equation (B.3) (Kreinin and Sidelnikova, 2001). These algorithms are fast and simple to implement, making them attractive for practitioners (see e.g. Bolder, 2022, pp. 421–422). They are also commonly included in embedding algorithm comparisons (see e.g. Inamura, 2006; Hughes and Werner, 2016; Dos Reis and Smith, 2018; Stokes, 2024).

We use the output of the DA method as the initial value for the EM algorithms and $L^2$-norm alternatives. Based on numerical experimentation, we conclude that varying this initial value does not significantly impact the performance of the embedding algorithms. Moreover, we consider a uniform initial distribution $\pi_k \equiv \frac{1}{K}$ for all algorithms.

### *3.1.2. Transition matrices*

Inspired by Kreinin and Sidelnikova (2001) and Hughes and Werner (2016), we consider the three transition matrices given in (Israel et al., 2001) and the four transition matrices in (J.P. Morgan, 1997). Similarly, inspired by Stokes (2024), we consider several transition matrices reported in (S&P Global, 2025): namely, their transition matrices in Tables 20.1–20.4, 21.1, 22.1–22.3, and 23.

We note that the considered empirical credit rating transition matrices all have a similar structure, as they tend to be diagonally dominant and have relatively small single-step absorption probabilities. This structure naturally arises whenever the time horizon of the transition matrix is relatively short compared to the stochastic system dynamics, such as with chronic diseases in health modeling or slowly degrading machines in systems engineering. We therefore view these matrices as representative for evaluating our methods in practical settings across different domains.

### *3.2. Results*

First, we assess the performance in preserving survival probabilities through the loss functions $\mathrm{Loss}_{\mathrm{KL}}^{\mathrm{Survival}}$ defined in Equation (3) and $\mathrm{Loss}_{L^2}^{\mathrm{Survival}}$ defined in Equation (6). Then, we evaluate the performance in fitting overall transition probabilities through the $n$-step snapshot $\mathrm{Loss}_{\mathrm{KL}}^{[\mathrm{D}_0,\mathrm{D}_n]}$ defined in Equation (2) for the cases $n = 1, 5, 10, 20$ years. We recall that the single-step case $n = 1$ corresponds to the $\mathrm{Loss}_{\mathrm{KL}}^{\mathrm{Standard}}$ defined in Equation (1).

The box-plots in Figure 1 visualize the relative performance distribution of the algorithms across the 16 transition matrices for each loss function. For better comparability across matrices, the results are normalized relative to the median-performing algorithm and expressed as a ratio relative to the median loss. The vertical baseline corresponding to a ratio of 1 represents the median performance for that matrix across all algorithms. This creates a straightforward visual interpretation, where boxes to the left (right) of the baseline perform better (worse) than the median. The loss ratios are plotted on a logarithmic scale, as they often span multiple orders of magnitude across the different algorithms.

Importantly, we observe that the algorithms consistently attain the lowest value for their targeted loss function across the 16 transition matrices, verifying that they work as intended.

*Preserving survival probabilities.* Importantly, the results show that our Enhanced EM algorithm outperforms existing methods in preserving survival probabilities, sometimes by multiple orders of magnitude. Notably, it also performs well for $\mathrm{Loss}_{L^2}^{\mathrm{Survival}}$, indicating robustness beyond the targeted loss function. The Standard EM algorithm performs best among all existing benchmarks.

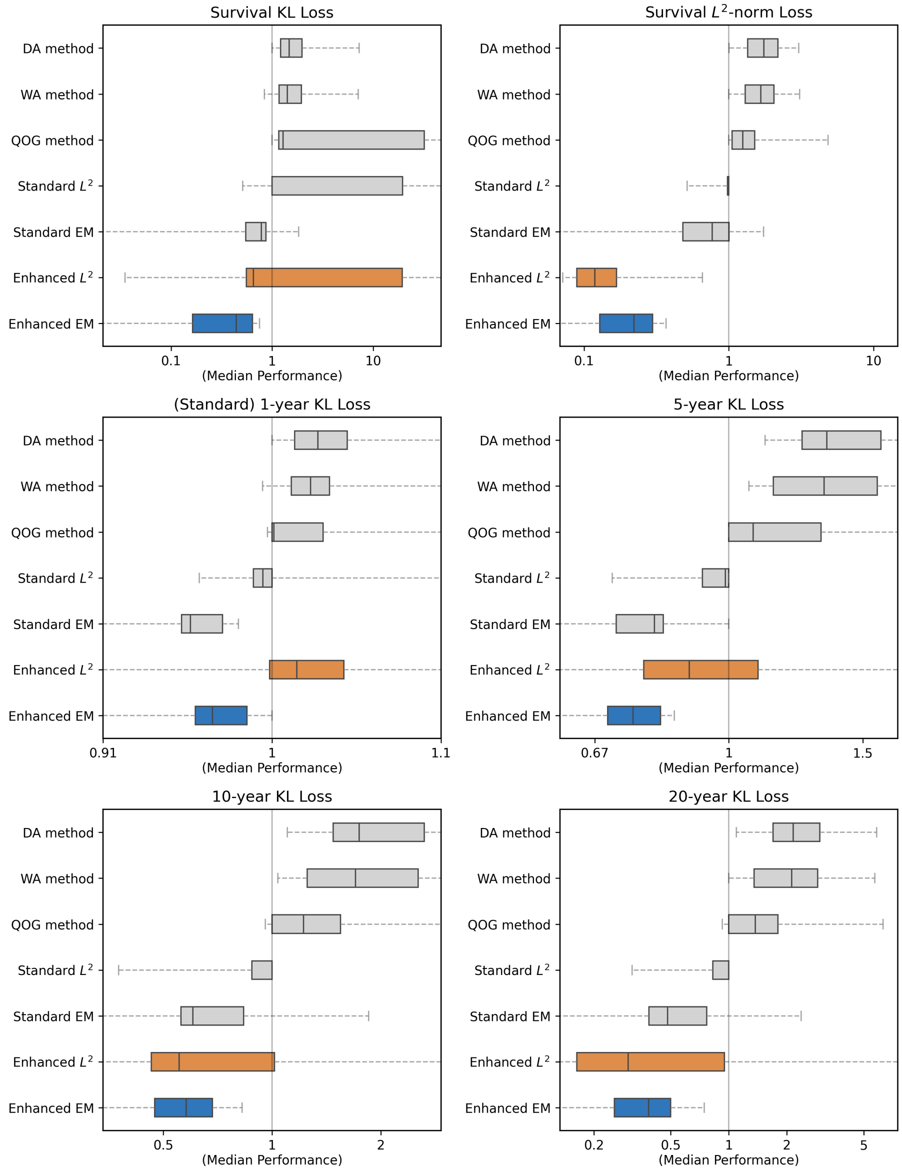


Figure 1: Relative performance distribution of embedding algorithms across 16 empirical credit rating transition matrices. The panels display box plots for six distinct loss functions: $\mathrm{Loss}_{\mathrm{KL}}^{\mathrm{Survival}}, \mathrm{Loss}_{L^2}^{\mathrm{Survival}}, \mathrm{Loss}_{\mathrm{KL}}^{\mathrm{Standard}}, \mathrm{Loss}_{\mathrm{KL}}^{[\mathrm{D}_0,\mathrm{D}_5]}, \mathrm{Loss}_{\mathrm{KL}}^{[\mathrm{D}_0,\mathrm{D}_{10}]}, \mathrm{Loss}_{\mathrm{KL}}^{[\mathrm{D}_0,\mathrm{D}_{20}]}$). The performance is expressed as a loss ratio relative to the median loss for each matrix and plotted on a logarithmic scale.

*Overall state transitions.* Additionally, the results show that our Enhanced EM algorithm also performs well in fitting overall state transitions. As expected, the existing Standard EM algorithm performs better for its target $\text{Loss}_{\text{KL}}^{\text{Standard}}$. Notably, our Enhanced EM algorithm performs better for overall transition dynamics corresponding to longer time horizons of $n = 5, 10, 20$ years, again indicating robustness beyond the targeted loss function.

## 4. Conclusion

We propose a framework for constructing embedding algorithms that preserve the probability distribution of a chosen target variable by minimizing KL divergence. We highlight a link with EM algorithms and demonstrate that several existing embedding algorithms can be considered special cases of our framework. As a motivating example, we use our framework to construct a new Enhanced EM algorithm designed for absorbing Markov chains. We demonstrate that it better preserves the probability distribution of survival times over long time horizons, while maintaining a balanced fit to overall transition dynamics.

Our work provides useful tools for constructing tailored loss functions and embedding algorithms. In particular, we derive new matrix-form expressions for the E-step of our EM algorithms and a closed-form matrix gradient for $L^2$-norm embedding loss functions.

## 5. Declarations

The author declares no conflicts of interest regarding this paper.

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## References

Albert, A., 1962. Estimating the infinitesimal generator of a continuous time, finite state Markov process. The Annals of Mathematical Statistics 33, 727–753.

Baake, M., Sumner, J., 2024. Embedding of Markov matrices for $d \leq 4$. Journal of Mathematical Biology 89, 23.

Bladt, M., Nielsen, B.F., 2017. Matrix-exponential distributions in applied probability. Springer.

Bladt, M., Sørensen, M., 2005. Statistical inference for discretely observed Markov jump processes. Journal of the Royal Statistical Society Series B: Statistical Methodology 67, 395–410.

Bladt, M., Sørensen, M., 2009. Efficient estimation of transition rates between credit ratings from observations at discrete time points. Quantitative Finance 9, 147–160.

Bolder, D.J., 2022. Modelling Economic Capital. Springer.

Camellini, F., Franchini, G., Prato, M., 2025. Computing $p$th root of a transition matrix with a deep unsupervised learning approach. International Journal of Computer Mathematics 102, 577–594.

Carette, P., Guerry, M.A., 2025. The Markov chain embedding problem in a one-jump setting. Journal of Applied Probability 62, 674–696.

Casanellas, M., Fernández-Sánchez, J., Roca-Lacostena, J., 2023. The embedding problem for Markov matrices. Publicacions Matemàtiques 67, 411–445.

Davies, E., 2010. Embeddable Markov Matrices. Electronic Journal of Probability 15, 1474–1486.

Dempster, A.P., Laird, N.M., Rubin, D.B., 1977. Maximum likelihood from incomplete data via the EM algorithm. Journal of the Royal Statistical Society. Series B, Methodological 39, 1–38.

Dos Reis, G., Smith, G., 2018. Robust and consistent estimation of generators in credit risk. Quantitative Finance 18, 983–1001.

Durastante, F., Meini, B., 2024. Stochastic $p$th root approximation of a stochastic matrix: A Riemannian optimization approach. SIAM Journal on Matrix Analysis and Applications 45, 875–904.

Elfving, G., 1937. Zur Theorie der Markoffschen Ketten. Acta Societatis Scientiarum Fennicae. Series A, Societas scientiarum Fennica.

Gzyl, H., Mayoral, S., 2025. Determining a credit transition matrix from cumulative default probabilities. An entropy minimization approach. Computational Economics 66, 1–17.

He, Q.M., Gunn, E., 2003. A note on the stochastic roots of stochastic matrices. Journal of Systems Science and Systems Engineering 12, 210–223.

Higham, N.J., 1989. Matrix nearness problems and applications. Applications of Matrix Theory 22, 1–27.

Hobolth, A., Jensen, J.L., 2005. Statistical inference in evolutionary models of DNA sequences via the EM algorithm. Statistical Applications in Genetics and Molecular Biology 4, article 18.

Hobolth, A., Jensen, J.L., 2011. Summary statistics for endpoint-conditioned continuous-time markov chains. Journal of Applied Probability 48, 911–924.

Hughes, M., Werner, R., 2016. Choosing Markovian credit migration matrices by nonlinear optimization. Risks 4, 31.

Inamura, Y., 2006. Estimating continuous time transition matrices from discretely observed data. Technical Report 06-E-7. Bank of Japan.

Israel, R.B., Rosenthal, J.S., Wei, J.Z., 2001. Finding generators for Markov chains via empirical transition matrices, with applications to credit ratings. Mathematical Finance 11, 245–265.

Jarrow, R.A., Lando, D., Turnbull, S.M., 1997. A Markov model for the term structure of credit risk spreads. The Review of Financial Studies 10, 481–523.

Joshi, P., Šmigoc, H., 2025. Arbitrarily finely divisible stochastic matrices. Linear Algebra and its Applications 722, 125–153.

J.P. Morgan, 1997. CreditMetrics™—Technical Document.

Kalbfleisch, J.D., Lawless, J.F., 1985. The analysis of panel data under a Markov assumption. Journal of the American Statistical Association 80, 863–871.
Kingman, J.F.C., 1962. The imbedding problem for finite Markov chains. Zeitschrift für Wahrscheinlichkeitstheorie und verwandte Gebiete 1, 14–24.
Kreinin, A., Sidelnikova, M., 2001. Regularization algorithms for transition matrices. Algo Research Quarterly 4, 23–40.
Lin, L., 2011. Roots of stochastic matrices and fractional matrix powers. PhD Thesis, University of Manchester.
Magnus, J.R., Neudecker, H., 2019. Matrix differential calculus with applications in statistics and econometrics. John Wiley & Sons.
McLachlan, G.J., Krishnan, T., 2008. The EM algorithm and extensions. John Wiley & Sons.
Olsson, M., 1996. Estimation of phase-type distributions from censored data. Scandinavian Journal of Statistics 23, 443–460.
S&P Global, 2025. Default, Transition, and Recovery: 2024 Annual Global Sovereign Default And Rating Transition Study.
Stokes, J.R., 2024. Infinitesimal Generator Estimation with an Application to Credit Risk. The Journal of Fixed Income 34.
Tataru, P., Hobolth, A., 2011. Comparison of methods for calculating conditional expectations of sufficient statistics for continuous time Markov chains. BMC Bioinformatics 12, 465.
Van-Brunt, A., 2018. Infinitely divisible nonnegative matrices, $M$-matrices, and the embedding problem for finite state stationary Markov chains. Linear Algebra and its Applications 541, 163–176.
Van Loan, C., 1978. Computing integrals involving the matrix exponential. IEEE Transactions on Automatic Control 23, 395–404.
Virtanen, P., Gommers, R., Oliphant, T.E., Haberland, M., Reddy, T., Cournapeau, D., Burovski, E., Peterson, P., Weckesser, W., Bright, J., et al., 2020. SciPy 1.0: fundamental algorithms for scientific computing in Python. Nature methods 17, 261–272.
Wu, C.F.J., 1983. On the convergence properties of the EM algorithm. The Annals of Statistics 11, 95–103.

## Appendix A. Appendix

This section outlines several technical derivations that were omitted from the main text. Many of the results are given in terms of the matrix exponential integral $\mathcal{I}^{(n)}(A) := \int_0^n e^{Q(n-t)} A e^{Qt} \mathrm{d}t$, which attains a well-known closed-form solution as the upper-diagonal block of the matrix exponential of an auxiliary block-matrix (Van Loan, 1978). That is

$$e^{Hn} = \begin{bmatrix} e^{Qn} & \mathcal{I}^{(n)}(A) \\ 0 & e^{Qn} \end{bmatrix}, \quad \text{where } H := \begin{bmatrix} Q & A \\ 0 & Q \end{bmatrix}.$$

Possible alternative techniques for computing matrix exponential integrals include uniformization and eigendecomposition of the transition matrix (see e.g. Hobolth and Jensen, 2011; Tataru and Hobolth, 2011).

*Appendix A.1. Expectation-Maximization*

Proposition 1 and 2 provide the matrix-form expressions for the expectation step of the EM algorithm for different cases of the target variable $Y$. In the proofs, we use the well-known expressions for expectations of end-conditioned sufficient statistics of CTMCs (see e.g. Hobolth and Jensen, 2005). Writing these element-wise expressions as a matrix-vector product yields

$$\begin{aligned} \mathbb{E}^Q\left[T_k \mathbb{1}\{\mathrm{D}_n = b\} \mid \mathrm{D}_0 = a\right] &= u_k^\top \mathcal{I}^{(n)}(u_b u_a^\top) u_k, \\ \mathbb{E}^Q\left[J_{kl} \mathbb{1}\{\mathrm{D}_n = b\} \mid \mathrm{D}_0 = a\right] &= Q_{kl} u_l^\top \mathcal{I}^{(n)}(u_b u_a^\top) u_k. \end{aligned} \tag{A.1}$$

It should be noted that the diagonal elements of $J$ always equal 0, as we do not count "jumps" that stay within the same state. Although these diagonal elements are not used in the EM algorithm, we enforce this zero constraint through element-wise multiplication with matrix $\tilde{Q}$, which equals $Q$ except for having zeros on the diagonal. Moreover, it is implicitly assumed that $Q_{kl} > 0$ for all $k \neq l$ such that $P_{kl} > 0$, so the KL divergence does not equal $\infty$. It is sufficient to start the EM algorithm with an initial $Q$ that satisfies this constraint.

**Proposition 1.** *For $Y = [\mathrm{D}_0, \mathrm{D}_n]$, it holds that*

$$\mathbb{E}^P\left[\mathbb{E}^Q\left[T \mid Y\right]\right] = \mathrm{diag}(Z), \quad \mathbb{E}^P\left[\mathbb{E}^Q\left[J \mid Y\right]\right] = \tilde{Q} \odot Z^\top,$$

*where $Z = \mathcal{I}^{(n)}(A)$, and matrix $A$ has elements $A_{ba} = \pi_a P^n_{ab} / [e^{Qn}]_{ab}$.*

*Proof.* Using the result in (A.1), we obtain

$$\begin{aligned} \mathbb{E}^P\left[\mathbb{E}^Q\left[J_{kl} \mid \mathrm{D}_0, \mathrm{D}_n\right]\right] &= \sum_{a,b \in E} \pi_a \mathbb{P}^P_a[\mathrm{D}_n = b] \frac{\mathbb{E}^Q\left[J_{kl} \mathbb{1}\{\mathrm{D}_n = b\} \mid \mathrm{D}_0 = a\right]}{\mathbb{P}^Q_a[\mathrm{D}_n = b]} \\ &= Q_{kl} u_l^\top \mathcal{I}^{(n)}\left(\sum_{a,b \in E} \pi_a \frac{\mathbb{P}^P_a[\mathrm{D}_n = b]}{\mathbb{P}^Q_a[\mathrm{D}_n = b]} u_b u_a^\top\right) u_k. \end{aligned} \tag{A.2}$$

The matrix-form result follows from recognizing $A$ inside the parentheses and noting that $[Q \odot Z^\top]_{kl} = Q_{kl} u_l^\top Z u_k$. The result for $T$ follows analogously. □

The result in Proposition 2 is derived for the more general case of truncated survival times $\tau \wedge N = \min(\tau, N)$, which reduces to $\tau$ for the special case of $N = \infty$. For finite $N$, this approach yields a finite sum expression plus a correction term that provides a better approximation than naively truncating the infinite sum expression. This can be considered a technicality, as long as $N$ exceeds the time horizon of interest while accounting for rare state transitions. For example, we use $N = 200$ for the numerical results in Section 3.

**Proposition 2.** *For $Y = [\mathrm{D}(0), \tau \wedge N]$, it holds that*

$$\mathbb{E}^P\left[\mathbb{E}^Q\left[T \mid Y\right]\right] = \mathrm{diag}(Z), \quad \mathbb{E}^P\left[\mathbb{E}^Q\left[J \mid Y\right]\right] = \tilde{Q} \odot Z^\top,$$

*where* $Z = \mathcal{I}^{(N-1)}\left(\bar{A}\right) + \sum_{n=1}^{N} \left[\mathcal{I}^{(n)}\left(A^{(n)}\right) - \mathcal{I}^{(n-1)}\left(A^{(n)}\right)\right]$, *and matrices* $\bar{A}, A^{(n)}$ *have elements* $\bar{A}_{ba} = \pi_a \frac{\mathbb{P}_a^P[\tau>N]}{\mathbb{P}_a^Q[\tau>N]} \mathbb{1}\{b \neq K\}$ *and* $A_{ba}^{(n)} = \pi_a \frac{\mathbb{P}_a^P[\tau=N]}{\mathbb{P}_a^Q[\tau=N]} \mathbb{1}\{b = K\}$.

*Proof.* Note that $\{\tau = n\} = \{\tau \leq n\} \setminus \{\tau \leq n-1\}$ and $\{\tau \leq n\} = \{\mathrm{D}_n = K\}$, so that $\mathbb{1}\{\tau = n\} = \mathbb{1}\{\mathrm{D}_n = K\} - \mathbb{1}\{\mathrm{D}_{n-1} = K\}$ holds for the absorbing state $K$. Using the result in (A.1), we thus obtain

$$\mathbb{E}^Q\left[J_{kl}\mathbb{1}\{\tau = n\} \mid \mathrm{D}_0 = a\right] = Q_{kl} u_l^\top \left[\mathcal{I}^{(n)}(u_K u_a^\top) - \mathcal{I}^{(n-1)}(u_K u_a^\top)\right] u_k. \quad \text{(A.3)}$$

Similarly, by noting that $\mathbb{1}\{\tau > n\} = \mathbb{1}\{\mathrm{D}_n \neq K\} = \sum_{b \neq K} \mathbb{1}\{\mathrm{D}_{n-1} = b\}$ and $\sum_{b\neq K} u_b = \mathbb{1} - u_K$, we obtain

$$\mathbb{E}^Q\left[J_{kl}\mathbb{1}\{\tau > n\} \mid \mathrm{D}_0 = a\right] = Q_{kl} u_l^\top \left[\mathcal{I}^{(n)}\left((\mathbb{1} - u_K)u_a^\top\right)\right] u_k. \quad \text{(A.4)}$$

Moreover, analogous to the decomposition in Equation (A.2), we have

$$\begin{aligned}
\mathbb{E}^P\left[\mathbb{E}^Q\left[J_{kl} \mid \mathrm{D}_0, \tau \wedge N\right]\right] &= \sum_{n=1}^{N} \sum_{a\in E} \pi_a \mathbb{P}_a^P[\tau \wedge N = n] \mathbb{E}^Q\left[J_{kl} \mid \mathrm{D}_0 = a, \tau \wedge N = n\right] \\
&= \sum_{a \in E} \pi_a \Bigg[ \sum_{n=1}^{N} \mathbb{P}_a^P[\tau = n] \frac{\mathbb{E}^Q\left[J_{kl} \mid \mathrm{D}_0 = a, \tau = n\right]}{\mathbb{P}_a^Q[\tau = n]} \\
&\qquad + \mathbb{P}_a^P[\tau > N] \frac{\mathbb{E}^Q\left[J_{kl} \mid \mathrm{D}_0 = a, \tau > N\right]}{\mathbb{P}_a^Q[\tau > N]} \Bigg].
\end{aligned}$$

The final result follows from substituting in Equations (A.3) and (A.4), moving terms inside the integrals and recognizing the matrix-form as in the proof of Proposition 1. The result for $T$ follows analogously. □

*Appendix A.2. General $L^2$-norm loss and matrix gradient.*

We consider a general $L^2$-norm loss function of the form

$$\mathrm{Loss}_{L^2}^{\mathrm{GEN}}(Q, P) = \sum_{n \in \mathcal{I}} \left\| \left(A\left(e^{Qn} - P^n\right)B\right) \odot W \right\|_{L^2}^2, \quad \text{(A.5)}$$

where $\mathcal{I} \subset \mathbb{N}$ is a set of positive integers, $A$ is a $p \times K$ matrix, $B$ is a $K \times q$ matrix, $W$ is a $p \times q$ matrix with non-negative entries, and $\odot$ represents the element-wise Hadamard product. Intuitively, $\mathcal{I}$ represents the time periods of interest, $A$ and $B$ represent linear transformations of transition probabilities, and $W$ represents the weights of state transitions.

The commonly considered $\mathrm{Loss}_{L^2}^{\mathrm{Standard}}(Q, P)$ in Equation (5) is a special case that corresponds to $\mathcal{I} = \{1\}$, $A = B = \mathrm{Id}_K$, and $W = \mathbb{1}\mathbb{1}^\top$. Similarly, it can be shown that the survival probability term structure loss in Equation (6) corresponds to $A = \mathrm{Id}_K$, $B = u_K$, a weights vector with elements $W_k \equiv \sqrt{\pi_k}$, and $\mathcal{I} = \{1, \ldots, N\}$. The result in Proposition 3 can be used for gradient-based optimization of these loss functions.

**Proposition 3.** *Let* $\text{Loss}_{L^2}^{\text{GEN}}$ *be defined as in Equation* (A.5)*, then the matrix gradient with elements* $\left[\frac{\mathrm{d}\text{Loss}_{L^2}^{\text{GEN}}(Q,P)}{\mathrm{d}Q}\right]_{ij} = \frac{\mathrm{d}\text{Loss}_{L^2}^{\text{GEN}}(Q,P)}{\mathrm{d}Q_{ij}}$*, is given by*

$$\frac{\mathrm{d}\text{Loss}_{L^2}^{\text{GEN}}(Q,P)}{\mathrm{d}Q} = 2\sum_{n\in\mathcal{I}} \left[\mathcal{I}^{(n)}\Big(B(\mathcal{E}^{(n)} \odot W)^\top A\Big)\right]^\top$$

*where* $\mathcal{E}^{(n)} := \left(A\left(e^{Qn} - P^n\right)B\right) \odot W$.

*Proof.* We prove the result by applying standard differential matrix calculus techniques (see Magnus and Neudecker, 2019). In particular, we repeatedly use the linearity of sums, differentials, trace operators, Hadamard products, matrix products and integrals. Since $\text{Loss}_{L^2}^{\text{GEN}} = \sum_{n\in\mathcal{I}} \text{Tr}\big({\mathcal{E}^{(n)}}^\top \mathcal{E}^{(n)}\big)$, it follows that

$$\begin{aligned}\mathrm{d}\text{Loss}_{L^2}^{\text{GEN}} = 2\sum_{n\in\mathcal{I}} \text{Tr}\big({\mathcal{E}^{(n)}}^\top \mathrm{d}\mathcal{E}^{(n)}\big) &= 2\sum_{n\in\mathcal{I}} \text{Tr}\left({\mathcal{E}^{(n)}}^\top \left[\big(A\ \mathrm{d}\left[e^{Qn}\right]B\big) \odot W\right]\right)\\ &= 2\sum_{n\in\mathcal{I}} \text{Tr}\left(\left[\mathcal{E}^{(n)} \odot W\right]^\top A\ \mathrm{d}\left[e^{Qn}\right]B\right).\end{aligned}$$

Using $\mathrm{d}\left[e^{Qn}\right] = \int_0^n e^{Qt}\mathrm{d}Q e^{Q(n-t)}\mathrm{d}t$, the trace cycle property and the aforementioned linearity, we obtain

$$\mathrm{d}\text{Loss}_{L^2}^{\text{GEN}} = 2\sum_{n\in\mathcal{I}} \text{Tr}\left(\int_0^n e^{Q(n-t)} B(\mathcal{E} \odot W)^\top A\ e^{Qt}\mathrm{d}t\ \mathrm{d}Q\right).$$

Finally, the result follows from recognizing the matrix exponential integral and applying the matrix gradient identification theorem (Magnus and Neudecker, 2019, p. 176), which states that $\mathrm{d}f(Q) = \text{Tr}\left(C\mathrm{d}Q\right) \iff \frac{\mathrm{d}f(Q)}{\mathrm{d}Q} = C^\top$. □

## Appendix B. Algorithms

This section outlines some algorithm details that were omitted from the main text: The outline of the EM algorithm described in Section 2.2.2 and more information about the regularization benchmarks considered in Section 3.

*Appendix B.1. EM algorithm*

**Algorithm 1** EM algorithm for continuous-time embedding.

**Input:** Transition matrix $P$, initial generator $Q^{(\text{initial})}$ and tolerance $\varepsilon > 0$.
**Output:** Embedded generator $Q$.
**Computation:**
Set $Q^{(\text{new})} \leftarrow Q^{(\text{initial})}$.
**repeat**
  Set $Q^{(\text{old})} \leftarrow Q^{(\text{new})}$.

  **Expectation step:**
  Compute $\widehat{J} \leftarrow \mathbb{E}^P \left[ \mathbb{E}^{Q^{(\text{old})}} [J \mid Y] \right],\ \widehat{T} \leftarrow \mathbb{E}^P \left[ \mathbb{E}^{Q^{(\text{old})}} [T \mid Y] \right].$

  **Maximization step:**
  Update $Q^{(\text{new})}_{kl} \leftarrow \widehat{J}_{kl}/\widehat{T}_k,$ for all $k \neq l$.
  Compute $Q^{(\text{new})}_{kk} \leftarrow -\sum_{l \neq k} Q^{(\text{new})}_{kl},$ for all $k \in E$.

**until** $\|Q^{(\text{new})} - Q^{(\text{old})}\| < \varepsilon$

$Q \leftarrow Q^{(\text{new})}$
**return** $Q$

*Appendix B.2. Regularization algorithms*

A real-valued matrix logarithm $\hat{Q}^{\log} := \log(P)$ provides an intuitive candidate for the continuous-time embedding of a transition matrix $P$. However, it may not satisfy the constraints of a generator matrix due to negative off-diagonal values. Regularization algorithms start with a non-valid candidate and modify it to obtain a valid generator matrix. For example, Israel et al. (2001) suggest the Diagonal Adjustment (DA) and Weighted Adjustment (WA) methods. The DA method removes negative off-diagonal values and adjusts the diagonal element to ensure that the rows still add up to 0

$$Q^{\text{DA}}_{kl} = \max(\hat{Q}^{\log}_{kl}, 0), \quad k \neq l; \quad Q^{\text{DA}}_{kk} = \hat{Q}^{\log}_{kk} + \sum_{l \neq k} \min(\hat{Q}^{\log}_{kl}, 0). \tag{B.1}$$

The WA method instead adjusts valid entries in proportion to their (absolute) size.

$$Q^{\text{WA}}_{kl} = \begin{cases} 0 & \text{if } k \neq l \text{ and } \hat{Q}^{\log} < 0 \\ \hat{Q}^{\log}_{kl} - B_k \left| \hat{Q}^{\log}_{kl} \right| / G_k & \text{otherwise if } G_k > 0 \\ \hat{Q}^{\log}_{kl} & \text{otherwise if } G_k = 0 \end{cases}, \tag{B.2}$$

where $G_k = \left| \hat{Q}^{\log}_{kk} \right| + \sum_{l \neq k} \max(\hat{Q}^{\log}_{kl}, 0)$ and $B_k = \sum_{l \neq k} \max(-\hat{Q}^{\log}_{kl}, 0)$. Although these methods were initially proposed as ad hoc approaches, Davies (2010) later proved that the DA method yields the generator that is closest to

$\hat{Q}^{\log}$ for a specific choice of matrix norm. More specifically, they prove that $Q^{\text{DA}} = \arg\min_{Q\in\mathcal{Q}} \left\|Q - \hat{Q}^{\log}\right\|_{\infty}$, where the matrix norm $\|\cdot\|_{\infty}$ is defined as $\|A\|_{\infty} := \max\{\|Av\|_{\infty} : \|v\|_{\infty} \leq 1\} = \max_{1\leq k\leq K}\left\{\sum_{l=1}^{K} |A_{kl}|\right\}$.

Kreinin and Sidelnikova (2001) instead propose an algorithm to find the Quasi-Optimizing Generator $Q^{\text{QOG}}$ that minimizes the $L^2$-norm distance relative to $\hat{Q}^{\log}$, i.e.

$$Q^{\text{QOG}} = \underset{Q\in\mathcal{Q}}{\arg\min} \left\|Q - \hat{Q}^{\log}\right\|_{L^2}^2, \tag{B.3}$$

Kreinin and Sidelnikova (2001) derive an algorithm that essentially consists of a row-by-row projection of $\hat{Q}^{\log}$ onto the simplex that represents the linear constraints for a valid generator matrix. Hughes and Werner (2016) consider the same minimization problem, but instead use a numerical non-linear optimization approach, which we also adopt. In our implementation, we compute the principal matrix logarithm using the `scipy.linalg.logm` function from the SciPy Python library Virtanen et al. (2020).